SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE

Rakshak Edge

A private, on-device social-engineering firewall for Snapdragon-powered PCs

PAUSE THE SCAM BEFORE THE TAP

Hariharan C | September 2026

Executive summary

Rakshak Edge analyzes suspicious messages, payment requests, QR payloads, and links before a user acts. It detects urgency, impersonation, credential theft, risky URL structure, and Indian payment signals. It then explains the evidence and recommends a safer verification path. The default build works fully offline with no account, telemetry, or cloud upload.

> **The product protects the decision point: the few seconds before a person clicks a link, shares an OTP, or approves a payment.**

Problem

Social-engineering scams exploit context, urgency, and trusted identities. A link scanner alone misses scams such as digital-arrest calls or requests from a compromised contact. A cloud-only checker introduces another privacy decision because the user must upload the very message, account detail, or payment request they distrust.

Solution

Rakshak Edge accepts copied text from SMS, email, WhatsApp, a URL, or a QR decoder. It extracts security-relevant entities, scores visible manipulation patterns, and produces an evidence-first report. High-risk results begin with a clear stop action and direct the user to an independently trusted contact path.

Evaluation area	Evidence in this submission
Technical implementation	Working local API, entity extraction, URL analysis, calibrated rules, optional ONNX classifier adapter, tests, and benchmark harness
Use case and innovation	Pre-action safety for link and non-link scams, Indian payment vocabulary, multilingual signals, and explainable guidance
Deployment and accessibility	Zero-dependency demo, CPU fallback, QNN production path, keyboard support, high contrast, and reduced motion
Presentation and documentation	Pitch deck, brief, live demo script, architecture, threat model, deployment guide, and clean repository

Technical implementation

The architecture uses a hybrid pipeline because security decisions need both semantic context and traceable evidence.

1	**Input boundary**	Loopback-only API, bounded payload, no server-side persistence
2	**Transparent analysis**	URL structure, UPI handles, phone numbers, amounts, credential requests, authority claims, urgency, and coercion
3	**Semantic model**	Optional MobileBERT-class INT8 classifier for paraphrases and mixed language
4	**Evidence fusion**	Calibrated score, top reasons, extracted entities, and safe actions

Snapdragon optimization

The production model runs through ONNX Runtime's Qualcomm QNN Execution Provider on Snapdragon X Series. Static INT8 quantization reduces memory traffic. Fixed input shapes avoid graph recompilation. Tokenization and the transparent rule engine stay on CPU, while the transformer encoder targets the Hexagon NPU. The adapter falls back to CPU when QNN is unavailable, and the full deterministic product remains usable when no model is installed.

Default cloud calls 0	**Reference rules latency** ~0.06 ms median*	**Model precision target** INT8	**Target warm latency** <100 ms

*The reference figure covers the deterministic analyzer on the development machine. It does not claim NPU model performance. Target-device validation remains an explicit acceptance check.

Safety and responsibility

Rakshak Edge reports risk indicators and safer actions. It does not claim that a message is proven fraudulent or safe. The UI uses textContent for user-derived values, the server constrains static-file paths, and the content security policy limits scripts and connections to the local origin. The threat model documents identity verification, malware sandboxing, and network reputation as out of scope.

Product experience

The interface uses a direct three-step mental model: paste, understand, act safely. Demo scenarios cover KYC and OTP theft, digital arrest, job-deposit fraud, and a legitimate security warning. Each result shows measured local latency, top evidence, and one primary action.

Scenario	Expected evidence	Primary response
KYC + OTP	Credential request, false urgency, look-alike SBI domain, UPI payment	Stop. Contact the bank through the official app or known number.
Digital arrest	Authority claim, secrecy, call control, immediate transfer	End the call and verify with a trusted person or official channel.
Safe warning	Security-awareness context discounts quoted risk terms	No strong pattern found; continue to verify unexpected requests.

Run and evaluate

Start the product with **python app.py**, then open **http://127.0.0.1:8765**. Run the regression suite with **python -m unittest discover -s tests -v**. Run the reproducible performance harness with **python benchmarks/run_benchmark.py**.

Roadmap

The next milestone adds a consented multilingual training set, calibration across scam families, Windows share-target integration, and a QNN context cache validated on a Snapdragon-powered HP PC. A later release can add opt-in federation that shares model updates without raw message content.

Key references

1. ONNX Runtime, Qualcomm QNN Execution Provider: https://onnxruntime.ai/docs/execution-providers/QNN-ExecutionProvider.html
2. Qualcomm AI Hub: https://aihub.qualcomm.com/
3. Snapdragon X Elite product brief: https://docs.qualcomm.com/bundle/publicresource/87-71417-1_REV_D_Snapdragon_X_Elite_Product_Brief.pdf

Repository package

The folder contains the working application, tests, benchmarks, architecture, threat model, Snapdragon deployment guide, demo script, submission copy, pitch deck, and this brief. Add the public GitHub URL to the Unstop form after pushing the repository.